

COMPUTER NETWORKS

CE-SEM 5

VIVA QUESTIONS



A computer is the most incredible tool we've ever seen. It can be a writing tool, a communications center, a supercalculator, a planner, a filer and an artistic instrument all in one, just by being given new instructions, or software, to work from. There are no other tools that have the power and versatility of a computer.

— Steve Jobs —



Computer Networks

1. What is a Computer Network?

A computer network is a collection of interconnected computers and devices that can communicate with each other and share resources, information, and services.

2. What are the main goals of computer networks?

The main goals are Resource Sharing, High Reliability, Saving Money, and Powerful Communication.

3. What is Network Topology?

Network Topology refers to the physical or logical arrangement of computers and other devices in a network. It defines how devices are connected.

4. List different types of network topologies.

Common topologies are Bus, Star, Ring, Mesh, Tree, and Hybrid.

5. What is the OSI Model?

The OSI (Open Systems Interconnection) Model is a 7-layer framework used to understand how network protocols work and interact. It standardizes network communication.

6. Name the seven layers of the OSI model.

From top to bottom: Application, Presentation, Session, Transport, Network, Data Link, Physical.

7. What is the TCP/IP Model?

The TCP/IP model is a concise, 4-layer model that describes the core protocols used on the internet. It is more practical than the OSI model.

8. Name the four layers of the TCP/IP model.

Application, Transport, Internet, and Network Access.

9. Differentiate between the OSI and TCP/IP models.

OSI has 7 layers and is a theoretical model. TCP/IP has 4 layers and is a practical, implemented model used for the internet.

10. What is a Protocol?

A protocol is a set of rules that governs data communications. It defines what is communicated, how it is communicated, and when it is communicated.

11. What is an IP Address?

An IP Address is a unique numerical identifier assigned to each device connected to a network that uses the Internet Protocol for communication.

12. Differentiate between IPv4 and IPv6.

IPv4 is a 32-bit address (e.g., 192.168.1.1), while IPv6 is a 128-bit address (e.g., 2001:0db8:85a3::8a2e:0370:7334). IPv6 was developed to overcome the address exhaustion of IPv4.

13. What is Subnetting?

Subnetting is the process of dividing a large network into smaller, more manageable sub-networks to improve performance and security.

14. What is a MAC Address?

A MAC (Media Access Control) Address is a unique physical address assigned to a Network Interface Card (NIC) by the manufacturer. It operates at the Data Link layer.

15. Differentiate between a Switch and a Hub.

A Hub is a dumb device that broadcasts data to all ports. A Switch is an intelligent device that forwards data only to the specific destination device based on MAC address.

16. What is a Router?

A Router is a networking device that forwards data packets between different computer networks. It operates at the Network layer and uses IP addresses for routing.

17. What is a Gateway?

A Gateway is a device that connects two networks using different protocols. It acts as a "gate" between two networks.

18. What is the difference between TCP and UDP?

TCP is connection-oriented, reliable, and provides error checking (e.g., HTTP, FTP). UDP is connectionless, faster, but unreliable (e.g., DNS, VoIP).

19. What is HTTP?

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used for transmitting hypermedia documents, like HTML. It is the foundation of the World Wide Web.

20. What is HTTPS?

HTTPS is the secure version of HTTP. It uses SSL/TLS encryption to protect the data exchanged between a web browser and a website.

21. What is FTP?

FTP (File Transfer Protocol) is a standard network protocol used for the transfer of computer files between a client and a server on a computer network.

22. What is DNS?

DNS (Domain Name System) is the "phonebook of the Internet." It translates human-readable domain names (like google.com) into machine-readable IP addresses.

23. What is DHCP?

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and other network configuration parameters to devices on a network.

24. What is Ping?

Ping is a network administration utility used to test the reachability of a host on an IP network and to measure the round-trip time for messages.

25. What is a Firewall?

A Firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

26. What is Bandwidth?

Bandwidth is the maximum rate of data transfer across a given path. It is typically measured in bits per second (bps).

27. What is Latency?

Latency is the time delay in a network. It is the time it takes for a data packet to travel from the source to the destination.

28. What is a Socket?

A Socket is one endpoint of a two-way communication link between two programs running on the network. It is a combination of an IP address and a port number.

29. What is a Port Number?

A Port Number is a 16-bit number used to identify a specific process or application on a device to which network messages are forwarded.

30. What are Well-Known Ports?

Well-Known Ports are the ports ranging from 0 to 1023 and are assigned to common services (e.g., Port 80 for HTTP, Port 21 for FTP).

31. What is ARP?

ARP (Address Resolution Protocol) is used to map a network layer address (IP) to a data link layer address (MAC).

32. What is Routing?

Routing is the process of selecting a path for traffic in a network or between or across multiple networks.

33. What is a Routing Table?

A Routing Table is a data table stored in a router or a network host that lists the routes to particular network destinations.

34. What is ICMP?

ICMP (Internet Control Message Protocol) is used by network devices to send error messages and operational information. The ping command uses ICMP.

35. What is NAT?

NAT (Network Address Translation) is a method of remapping one IP address space into another by modifying network address information in the IP header of packets while they are in transit across a traffic routing device.

36. What is VLAN?

A VLAN (Virtual Local Area Network) is a logical subnetwork that can group together devices from different physical LANs. It improves security and reduces broadcast traffic.

37. What is the Data Link Layer?

The Data Link Layer is responsible for node-to-node data transfer and error correction from the Physical layer. It is divided into LLC and MAC sublayers.

38. What is the role of the Network Layer?

The Network Layer is responsible for packet forwarding including routing through intermediate routers and logical addressing (IP addressing).

39. What is the role of the Transport Layer?

The Transport Layer is responsible for end-to-end communication, error recovery, flow control, and ensuring complete data transfer. It provides services like TCP and UDP.



40. What is Flow Control?

Flow Control is the process of managing the rate of data transmission between two nodes to prevent a fast sender from overwhelming a slow receiver.

41. What is Error Control?

Error Control involves mechanisms for detecting and correcting errors that occur in the transmission of frames. Techniques include CRC and ARQ.

42. What is CSMA/CD?

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is a media access control method used in early Ethernet. It listens before transmitting and detects collisions.

43. What is a Three-Way Handshake?

A Three-Way Handshake is a method used in a TCP/IP network to create a connection between a client and a server. The steps are: SYN, SYN-ACK, ACK.

44. What is a Cryptography?

Cryptography is the practice and study of techniques for secure communication in the presence of third parties. It involves encryption and decryption.

45. Differentiate between Symmetric and Asymmetric Key Cryptography.

Symmetric cryptography uses the same key for encryption and decryption (e.g., AES). Asymmetric cryptography uses a public key for encryption and a private key for decryption (e.g., RSA).

46. What is a Digital Signature?

A Digital Signature is a mathematical scheme for verifying the authenticity of digital messages or documents. It ensures that the message was created by a known sender and was not altered.

47. What is a VPN?

A VPN (Virtual Private Network) extends a private network across a public network, enabling users to send and receive data as if their devices were directly connected to the private network.

48. What is Wi-Fi?

Wi-Fi is a family of wireless network protocols based on the IEEE 802.11 standard, commonly used for local area networking and internet access.

49. What is Bluetooth?

Bluetooth is a wireless technology standard used for exchanging data over short distances, creating Personal Area Networks (PANs).

50. What is Cloud Computing?

Cloud Computing is the on-demand availability of computer system resources, especially data storage and computing power, without direct active management by the user. It relies heavily on computer networks for delivery.